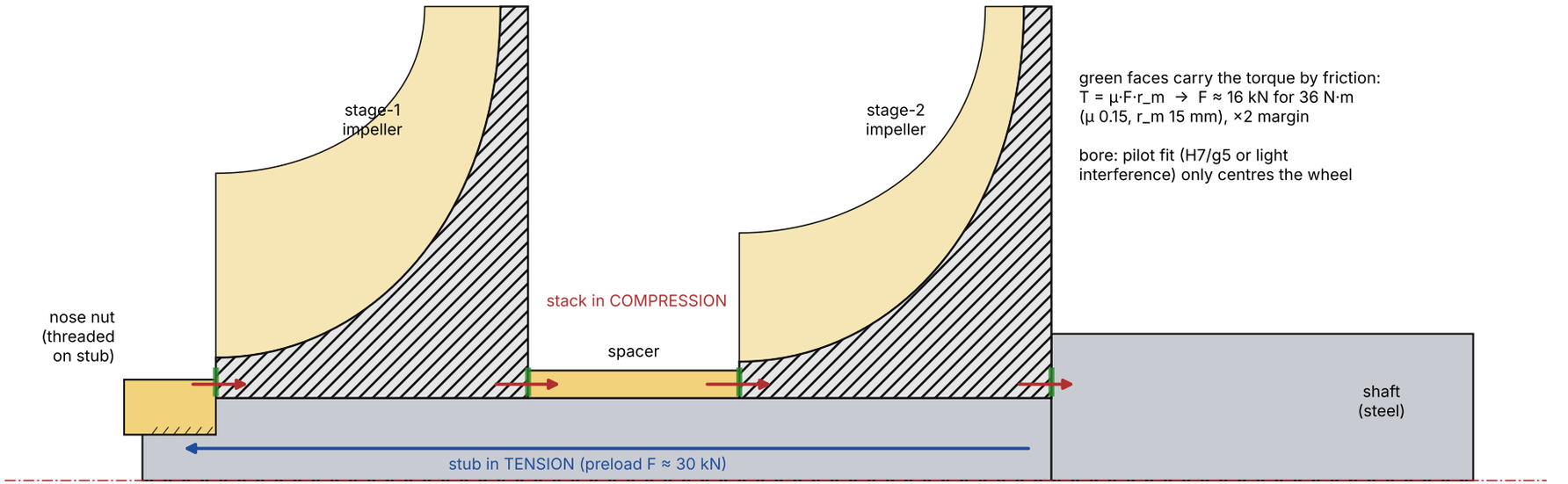
(a) Clamped stack: nut / tie bolt, torque by face friction

+ easy to assemble and service
+ balance stays repeatable
– needs flat, square faces (datum A)

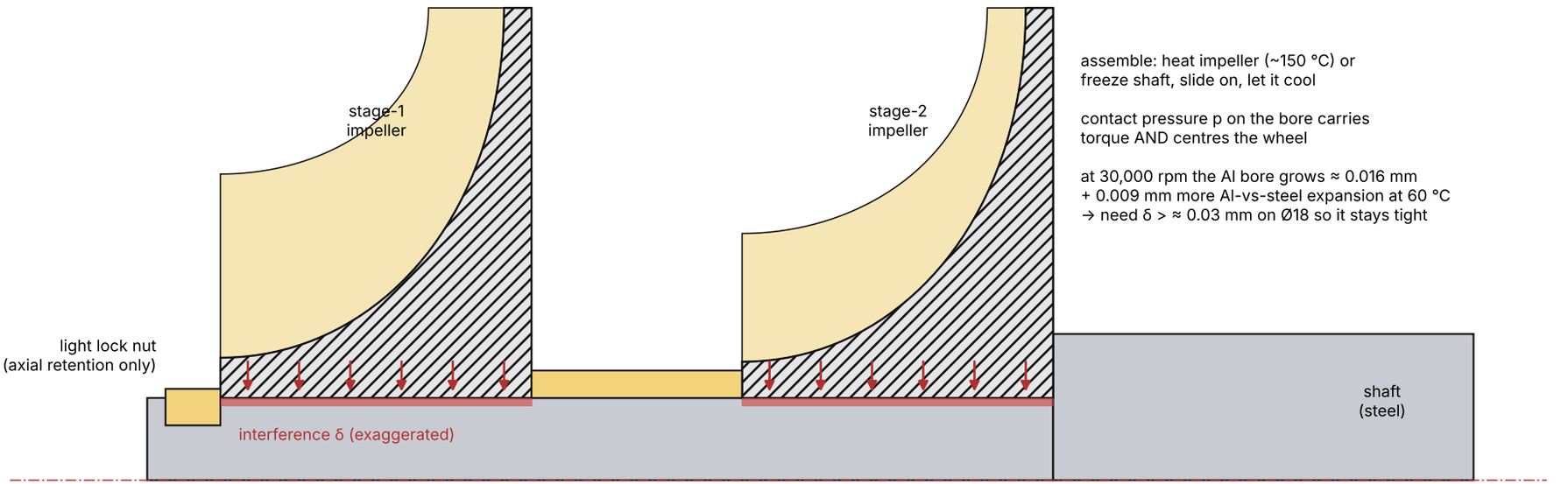


green faces carry the torque by friction:
$T = \mu \cdot F \cdot r_m \rightarrow F \approx 16$ kN for 36 N·m
(μ 0.15, r_m 15 mm), ×2 margin

bore: pilot fit (H7/g5 or light interference) only centres the wheel

(b) Shrink (interference) fit: torque by bore pressure

+ no torque on the faces, very stiff
– hard to remove; heating can change 7075-T6 temper above ~120 °C
– fit loosens with speed and temperature



assemble: heat impeller (~150 °C) or freeze shaft, slide on, let it cool

contact pressure p on the bore carries torque AND centres the wheel

at 30,000 rpm the Al bore grows ≈ 0.016 mm
+ 0.009 mm more Al-vs-steel expansion at 60 °C
→ need $\delta >$ ≈ 0.03 mm on Ø18 so it stays tight

Half-sections above the rotor axis, to scale (layout.py). Blades in yellow, hub body hatched. Study schematic - the real Turbocor mounting is not published.